

Boundary-Mode Topographic Cosmology: A Reduced-Response Bridge from Berger–Hopf Geometry to Late-Time Structure Growth

Norman P. Carberry
Independent Researcher

Working draft — September 2026

Abstract

We develop a late-time cosmological bridge between a scalar topographic effective field theory on compact spatial sections $S^3(R_H)$ and the constraint-reduced response architecture of the Berger–Hopf program. A closed-FLRW gauge audit changes the identification of the relevant infrared carrier: the first nonconstant scalar harmonic in our convention, $n = 1$, maps to the exceptional $N = 2$ sector of standard closed-universe perturbation theory and is nondynamical or pure gauge in the regular single-field covariant scalar sector. The first propagating scalar harmonic is therefore our $n = 2$ ($N = 3$ in the alternative convention). This does not remove the sky-dipole mechanism. A general degree-two scalar harmonic on S^3 contains observer-sky monopole, dipole, and quadrupole pieces, and a pure cross-term subspace produces an exact sky dipole proportional to $\sin[2\psi(z)] \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}$. The reduced Schur/Feshbach response can selectively soften this first physical harmonic while stiffening all propagating $n \geq 3$ modes when $1/15 < \ell^2 < 1/8$. Reference Branch R1 and its frozen numerical manifest are left unchanged because the background and high- n growth calculation do not depend on the identity of the curvature-scale carrier. The theoretical harmonic identification is amended prospectively before comparison with the targeted independent data. The resulting framework is falsifiable through curvature, the fixed dipole axis and radial envelope, compact-harmonic hierarchy, growth, and lensing tests.

1 Introduction

Late-time cosmological tensions and directional residuals motivate tests that are capable of separating local structure, survey geometry, and genuinely new large-scale dynamics. The scalar-topographic program introduced a compact hyperspherical parent $S^3(R_H)$ and a long-wavelength scalar response whose largest observational effect occurs at low redshift [1]. A later covariant formulation placed the topographic scalar in the cubic-Galileon sector of Horndeski theory, preserving second-order time evolution while allowing nontrivial late-time scalar–metric braiding.

The present work asks a narrower question: can the long-wavelength topographic response be represented as the large-scale branch of the same constraint-reduced response logic used in the Berger–Hopf program, without assuming that the full microscopic BHSM construction has already been compactified into cosmology? We answer this at the level of a controlled bridge model.

The central mathematical chain is

$$S[\Phi] \longrightarrow \hat{\mathcal{H}}_{\text{phys}} \longrightarrow \mathcal{K}_{\text{topo}} \longrightarrow \lambda_{\text{topo}} \longrightarrow T_n \longrightarrow \delta g_{\mu\nu} \longrightarrow \{D_L, H, f\sigma_8, \Phi + \Psi\}.$$

The distinction between established input and bridge extension is essential. The S^3 harmonic spectrum and cubic-Galileon background are treated as the cosmological starting point. The Schur/Feshbach reduction is an exact mathematical operation once a physical complement is specified. The identification of its low-energy coefficients with the topographic response is a bridge assumption to be tested, not a declaration that the complete microscopic BHSM action has already been reduced to four-dimensional cosmology.

The paper makes four main contributions. First, it standardizes the S^3 harmonic convention and shows how the observed sky dipole arises from the four-dimensional $n = 1$ eigenspace rather than by identifying a radial mode with an angular dipole. Second, it derives a reduced kernel in which a positive higher-spatial-gradient term can emerge after integrating out a stable complement. Third, it identifies a parameter interval in which the $n = 1$ response is selectively softened while all $n \geq 2$ harmonics are stiffened. Fourth, it gives an executable preregistration protocol for curvature, supernova anisotropy, growth, and lensing tests.

The claim boundary is therefore:

This work does not assume that the scalar-topographic cosmological field has already been derived from the complete microscopic Berger–Hopf action. Exact reductions are distinguished from cross-scale identifications and exploratory reference-branch assumptions. Prospective observables are frozen before comparison with designated independent data.

2 Hyperspherical Parent and Harmonic Convention

We use a closed FLRW geometry

$$ds^2 = -dt^2 + a^2(t)R_H^2 \left[d\psi^2 + \sin^2 \psi d\Omega_2^2 \right],$$

with $a_0 = 1$. Define

$$D_C(z) = c \int_0^z \frac{dz'}{H(z')}, \quad \psi(z) = \frac{D_C(z)}{R_H}.$$

Scalar harmonics satisfy

$$-\nabla_\gamma^2 Y_n = \frac{n(n+2)}{R_H^2} Y_n, \quad n = 0, 1, 2, \dots,$$

so the physical spatial eigenvalue is

$$p_n(a) = \frac{n(n+2)}{a^2 R_H^2}.$$

A common closed-universe convention uses $k^2 = N^2 - 1$, hence

$$\boxed{N = n + 1}.$$

2.1 Exceptional first nonconstant harmonic

For our $n = 1$, $k^2 = 3$ and $N = 2$. The degree-one scalar harmonics on the unit three-sphere obey

$$D_i D_j Q^{(1)} = -\gamma_{ij} Q^{(1)}.$$

Therefore

$$\left(D_i D_j + \frac{k^2}{3} \gamma_{ij} \right) Q^{(1)} = 0.$$

The scalar-derived trace-free metric harmonic is absent. For the regular generally covariant one-scalar sector, the remaining scalar gauge transformations and lapse/shift constraints leave no ordinary propagating scalar degree of freedom in this exceptional sector. Modern gauge-invariant closed-universe analyses likewise take the propagating scalar spectrum to begin at alternative-convention $N \geq 3$ [2, 3].

Accordingly, the minimal bridge no longer uses our $n = 1$ as the physical scalar-metric carrier. An additional boundary degree of freedom could evade this conclusion, but it is not required for the construction below.

2.2 First physical scalar harmonic and exact dipole subspace

The first propagating scalar harmonic is

$$\boxed{n_\star = 2}, \quad \boxed{N_\star = 3}, \quad p_2 = \frac{8}{a^2 R_H^2}.$$

Let the unit $S^3 \subset \mathbb{R}^4$ embedding coordinates be

$$X^A = (\cos \psi, \sin \psi \hat{n}^i).$$

Every real degree-two scalar harmonic can be written

$$T_2 = A_{AB} X^A X^B, \quad A_{AB} = A_{BA}, \quad \delta^{AB} A_{AB} = 0.$$

Writing $c = \cos \psi$, $s = \sin \psi$, $a_i = A_{4i}$, and

$$A_{ij} = S_{ij} - \frac{A_{44}}{3} \delta_{ij}, \quad S_i^i = 0,$$

gives the exact observer-shell decomposition

$$\boxed{T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j.}$$

These terms are respectively a sky monopole, dipole, and quadrupole. The pure cross-term subspace

$$A_{44} = 0, \quad S_{ij} = 0, \quad a_i \neq 0$$

therefore gives an exact sky dipole:

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}).}$$

Thus the gauge audit shifts the carrier to the first physical compact-space scalar mode while retaining an exact dipolar observer-sky sector.

3 Constraint-Reduced Response and the Topographic Kernel

Let $\hat{\mathcal{H}}$ be the physical Hessian after constraints and gauge directions have been removed. For a selected topographic branch P and physical complement Q , the exact Schur/Feshbach reduced operator is

$$\mathcal{K} = P \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q (Q \hat{\mathcal{H}} Q)^{-1} Q \hat{\mathcal{H}} \right] P.$$

For one scalar harmonic, write the quadratic block as

$$\hat{\mathcal{H}}_n = \begin{pmatrix} A(p) & C(p) \\ C^\dagger(p) & D(p) \end{pmatrix}.$$

The exact reduced stiffness is

$$K_{\text{red}}(p) = A(p) - C(p)D^{-1}(p)C^\dagger(p).$$

A minimal long-wavelength realization is

$$A(p) = Z_0 p, \quad D(p) = M_c^2 + c_c p, \quad C(p) = g\sqrt{p}.$$

Then

$$K_{\text{red}}(p) = Z_0 p - \frac{g^2 p}{M_c^2 + c_c p}.$$

For $c_c p \ll M_c^2$,

$$K_{\text{red}}(p) = \left(Z_0 - \frac{g^2}{M_c^2} \right) p + \frac{g^2 c_c}{M_c^4} p^2 + O(p^3).$$

Thus a positive p^2 stiffness can arise from a coupled second-order parent system after a stable complement is integrated out. In this interpretation the old static operator

$$-\nabla^2 + B\nabla^4$$

is a low-energy spatial derivative expansion, not evidence for a fundamental fourth-order time equation.

Define

$$\chi \equiv \frac{g^2}{M_c^2}, \quad L_c^2 \equiv \frac{c_c}{M_c^2}.$$

After canonical normalization of the leading spatial term, the reduced kernel may be written schematically as

$$K_{\text{topo}}(p) = m_T^2 + (c_s^2 - \chi)p + \chi L_c^2 p^2 + \dots.$$

The signs of the temporal kinetic term and of the full physical gradient matrix must be checked independently; a positive spatial p^2 coefficient by itself is not a complete ghost analysis.

4 Cubic-Galileon Scalar–Metric Bridge

The cosmological parent action is taken to be

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \nabla_\mu T \nabla^\mu T + \frac{1}{\Lambda^3} \square T (\nabla T)^2 - V(T) + \mathcal{L}_m \right].$$

Using

$$X \equiv -\frac{1}{2} \nabla_\mu T \nabla^\mu T,$$

this corresponds to

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0$$

in the convention $\mathcal{L}_3 = -G_3 \square T$.

The model belongs to the kinetic-gravity-braiding subclass of Horndeski theory [4, 5]. Since G_4 is constant and $G_5 = 0$,

$$M_*^2 = M_{\text{Pl}}^2, \quad \alpha_M = 0, \quad \alpha_T = 0.$$

The scalar–metric mixing is controlled by the braiding function

$$\alpha_B = \frac{2\dot{T}^3}{HM_{\text{Pl}}^2\Lambda^3},$$

and

$$\alpha_K = \frac{\dot{T}^2}{H^2 M_{\text{Pl}}^2} + 6\alpha_B.$$

In the Bellini–Sawicki organization of Horndeski perturbations the scalar kinetic combination is

$$D_{\text{kin}} = \alpha_K + \frac{3}{2}\alpha_B^2.$$

Their perturbation equations are formulated for a spatially flat background [5]. We therefore use $D_{\text{kin}} > 0$ below as an inherited kinetic diagnostic for the cubic-Galileon sector, not as a complete proof of perturbative stability for the present closed background plus the additional reduced spatial response. A full stability proof requires the constrained gauge-invariant kinetic and gradient matrices, including the exceptional lowest scalar sector identified above.

For numerical work we softly break shift symmetry with

$$V(T) = V_0 - \mu^3 T.$$

The scalar energy density and pressure are

$$\rho_T = \frac{1}{2}\dot{T}^2 + V(T) + \frac{6H\dot{T}^3}{\Lambda^3},$$

$$P_T = \frac{1}{2}\dot{T}^2 - V(T) - \frac{2\dot{T}^2\ddot{T}}{\Lambda^3}.$$

The shift current

$$J = \dot{T} + \frac{6H\dot{T}^2}{\Lambda^3}$$

obeys

$$\dot{J} + 3HJ = \mu^3.$$

These background expressions are the standard kinetic-gravity-braiding relations specialized to the present G_2, G_3 choice [4].

The closed-FLRW background equation is

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) = \rho_m + \rho_r + \rho_T.$$

The bridge does not require the bare Horndeski sound speed to become negative. Instead, the BHSM-inspired reduced response is used to lower the effective long-wavelength spatial stiffness while a positive p^2 term stiffens shorter wavelengths. This statement concerns the reduced spatial kernel; it is not, by itself, a complete closed-FLRW stability theorem.

5 BHSM-native transport of the physical topographic mode

The purpose of this section is to separate the ontology of the Berger–Hopf Standard Model (BHSM) mode from any particular four-dimensional effective-field-theory coordinate used to represent its late-time response. The distinction is important in the closed cosmological setting: the physical topographic carrier is not introduced as an additional free bulk scalar. Rather, it is an action-owned geometric/boundary mode whose physical dynamics are obtained only after the BHSM constraints and gauge directions have been removed.

5.1 Action ownership and physical tangent space

The BHSM construction used in this work treats modes as physical boundary-local fields acted on by sector projectors. The scalar/topographic sector is represented in the unified action candidate and admits an action-reduced mode functional together with a normalized profile-boundary boundary-value problem. These ingredients motivate the following physical tangent space at a background configuration $z_\star$:

$$T_{\text{phys}}(z_\star) = \ker DC(z_\star) \cap \mathcal{G}^\perp, \quad (1)$$

where $\mathcal{C}$ denotes the action-owned constraints and $\mathcal{G}$ the infinitesimal gauge directions. The corresponding projected Hessian is

$$\hat{\mathcal{H}} = P_{\text{phys}} D^2 \mathcal{L}_{\text{KKT}}(z_\star) P_{\text{phys}}. \quad (2)$$

If P_2 denotes the projector onto the first admissible physical scalar-topographic harmonic, the constraint-eliminated quadratic operator is the Schur–Feshbach reduction

$$\mathcal{K}_2 = P_2 \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q_2 \left(Q_2 \hat{\mathcal{H}} Q_2 \right)^{-1} Q_2 \hat{\mathcal{H}} \right] P_2, \quad (3)$$

whenever the eliminated block is invertible on the chosen physical domain. Here $Q_2 = 1 - P_2$ on the already gauge-reduced tangent space.

5.2 Cosmological topographic-mode theorem

Theorem 1 (action-owned $n = 2$ envelope transport). Assume that:

1. the cosmological parent envelope is the closed spatial manifold $\mathcal{E}_{\text{cos}} \simeq S^3(R_H)$;
2. the scalar/topographic perturbation belongs to the BHSM action-owned boundary-mode sector and is reduced to T_{phys} before physical propagation is assigned;
3. the homogeneous $n = 0$ component is absorbed into the background and the exceptional scalar $n = 1$ sector is removed by the regular closed-FLRW constraint/gauge quotient;
4. the reduced $n = 2$ kinetic form is nondegenerate and positive on the retained mode.

Then the first physical scalar-topographic carrier is an action-owned $n = 2$ envelope mode. It does not require the introduction of an additional fundamental bulk scalar degree of freedom. Its canonical momentum, when nonzero, is induced by the reduced action itself and is therefore conjugate data of the same geometric mode rather than a separately postulated field.

Proof. Let $e_2(x)$ be a normalized element of $P_2 T_{\text{phys}}$ and write the retained perturbation locally as

$$\delta\Phi_{\text{topo}}(t, x) = q_2(t)e_2(x). \quad (4)$$

Substitution into the gauge- and constraint-reduced quadratic action gives, up to linear source terms and a choice of time parametrization,

$$S_2^{(2)} = \frac{1}{2} \int dt a^3 \left[\mathcal{G}_2(t) (D_t q_2)^2 - \mathcal{M}_2^2(t) q_2^2 + 2 \mathcal{J}_2(t) q_2 \right]. \quad (5)$$

The coefficients are the e_2 matrix elements of the reduced operator (3); explicitly, $\mathcal{G}_2$ is the kinetic quadratic form, $\mathcal{M}_2^2$ the reduced spatial/restoring form, and $\mathcal{J}_2$ the projection of the action-owned source onto e_2 . The canonical momentum is therefore

$$\Pi_2 = a^3 \mathcal{G}_2 D_t q_2, \quad (6)$$

which is generated by the same reduced action. Variation of (5) yields

$$D_t (a^3 \mathcal{G}_2 D_t q_2) + a^3 \mathcal{M}_2^2 q_2 = a^3 \mathcal{J}_2. \quad (7)$$

Thus the dynamical phase-space pair (q_2, Π_2) is the reduced description of one action-owned envelope mode. No second ontological scalar is created by introducing an effective coordinate for that mode. $\square$

5.3 Relation to the exact closed-Horndeski kernel

The closed-Horndeski calculation of Sec. 6 is used here as a late-time effective response representation of the same reduced mode, not as the microscopic definition of the BHSM topographic ontology. In the Bellini–Sawicki convention used in that section, the exact physical- $n = 2$ kinetic coefficient is

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1 - \alpha_B^{\text{BS}}/2)^2}. \quad (8)$$

For $D_{\text{kin}} > 0$ and nonsingular braiding denominator this coefficient is positive. Equation (8) therefore supplies an effective kinetic matching condition for assumption 4 of Theorem 1 on the reference branch.

The Stückelberg variable π used in the EFT equations is accordingly treated as a coordinate representative of the retained response of q_2 . It is not counted as an additional BHSM mode. A change of EFT normalization amounts to an invertible local redefinition of the same reduced coordinate.

5.4 Exact observer dipole inside the physical $n = 2$ sector

A degree-two harmonic on S^3 may be written

$$T_2 = A_{AB} X^A X^B, \quad A_A{}^A = 0. \quad (9)$$

Relative to an observer shell,

$$T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j. \quad (10)$$

The pure A_{4i} subspace is therefore

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})}, \quad (11)$$

so the observed dipolar angular dependence is embedded exactly in the first physical scalar-topographic harmonic.

5.5 Role of the curved EFT Bianchi audit

The curvature-aware EFT system of the following sections remains a valuable cross-check of a chosen four-dimensional response representation. It is not used as the ontological definition of the BHSM mode and it is not allowed to introduce an additional independent scalar degree of freedom.

At the present stage, the literal curvature-aware 00 , $0i$, spatial-trace, and scalar equations and the reference background have been independently transcription-audited. The fully integrated curved-EFT differential-algebraic representation nevertheless retains a nonzero constraint-preservation defect, while the consistently flattened system closes to numerical precision. We therefore classify that calculation as an *EFT compatibility audit* rather than as the defining transport theorem.

The defining evolution statement for the bridge is Eq. (7): propagation is generated by the action-reduced physical $n = 2$ operator. A future exact identification of the curved EFT transfer map with $\mathcal{K}_2$ must reproduce this reduced phase space without duplicating degrees of freedom.

5.6 Claim boundary

Theorem 1 establishes the mode ownership and the form of its reduced transport law conditional on the BHSM action-owned scalar/topographic sector, the physical quotient, and positivity of the retained kinetic form. It does not by itself determine an absolute normalization for $\mathcal{M}_2^2$, $\mathcal{J}_2$, or the observable transfer coefficient. Those quantities are supplied, where stated, by the frozen reference branch or by explicit observational calibration and are labelled accordingly.

Consequently, the exact $n = 2$ dipole geometry, the physical-harmonic selection, and the positive effective kinetic matching are structural results. Reference growth curves and other quantities that require a particular EFT transfer realization remain reference-branch predictions rather than framework-wide theorems.

6 Exact Closed-Horndeski Kernel for the First Physical Scalar Mode

The previous section uses the Bellini–Sawicki functions to organize the cubic-Galileon background. For the curvature-scale mode, however, the flat-FLRW perturbation formulas are not sufficient. We therefore specialize the non-flat Horndeski quadratic action derived by Akama and Kobayashi [6].

For

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0,$$

their tensor coefficients reduce exactly to

$$\mathcal{F}_T = \mathcal{G}_T = M_{\text{Pl}}^2.$$

Because G_4 is constant and $G_5 = 0$, the explicit curvature contributions to their Θ and Σ vanish. The remaining coefficients are

$$\Theta = HM_{\text{Pl}}^2 - \frac{\dot{T}^3}{\Lambda^3}$$

and

$$\Sigma = \frac{1}{2}\dot{T}^2 + \frac{12H\dot{T}^3}{\Lambda^3} - 3M_{\text{Pl}}^2H^2.$$

Equivalently,

$$\Theta = HM_{\text{Pl}}^2 \left(1 - \frac{\alpha_B}{2}\right).$$

Define

$$r(t) \equiv \frac{M_{\text{Pl}}^2 \Sigma}{\Theta^2}, \quad \kappa(t) \equiv \frac{K}{a^2}.$$

The first physical scalar harmonic has alternative-convention index $N = 3$, so

$$k^2 = K(N^2 - 1) = 8K.$$

The two Akama–Kobayashi scalar constraints then reduce to

$$\Sigma \delta n + 8\Theta \kappa \chi + 3\Theta \dot{\zeta} + 5M_{\text{Pl}}^2 \kappa \zeta = 0,$$

$$\Theta \delta n - M_{\text{Pl}}^2 \dot{\zeta} - M_{\text{Pl}}^2 \kappa \chi = 0.$$

They can be solved algebraically:

$$\delta n = \frac{5M_{\text{Pl}}^2}{\Theta(r+8)} \left[\dot{\zeta} - \frac{M_{\text{Pl}}^2 \kappa}{\Theta} \zeta \right]$$

and

$$\chi = -\frac{r+3}{\kappa(r+8)} \dot{\zeta} - \frac{5M_{\text{Pl}}^2}{\Theta(r+8)} \zeta.$$

The reduced kinetic coefficient of this $n = 2$ mode is therefore

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5(r+3)}{r+8}.$$

Using the cubic-Galileon α -functions gives the exact identity

$$r+3 = \frac{D_{\text{kin}}}{2(1 - \alpha_B/2)^2},$$

so that

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1 - \alpha_B/2)^2}.$$

Thus, provided $\Theta \neq 0$, positivity of the previously computed D_{kin} is sufficient for positivity of the pure gravity–scalar kinetic coefficient of the first physical closed-universe scalar harmonic.

For completeness, the corresponding non-flat Horndeski gradient coefficient is

$$\mathcal{F}_{S,2} = \frac{1}{a} \frac{d}{dt} \left[\frac{5}{r+8} \left(\frac{aM_{\text{Pl}}^4}{\Theta} \right) \right] - M_{\text{Pl}}^2 + \frac{5}{r+8} \frac{M_{\text{Pl}}^6}{\Theta^2} \frac{K}{a^2}.$$

The repository evaluates this expression as a gravity–scalar diagnostic only. The late-time R1 background contains matter and radiation, whose perturbations introduce additional dynamical variables and source terms. Consequently, $\mathcal{F}_{S,2}$ is not promoted here to a complete matter-era stability or light-cone transfer result.

This section closes an important part of the bridge: the first physical harmonic has an explicit constrained scalar–metric kernel rather than only a geometric identification. The remaining observable calculation is the coupled matter/radiation plus topographic-response system and its gauge-invariant transformation to the metric and velocity variables entering distance, growth, and lensing observables.

7 Matter-Sourced Constraint Transfer at $n=2$

The Round-3 kernel established the pure gravity–scalar constrained dynamics of the first physical closed-universe harmonic. The next step is to introduce minimally coupled matter sources without importing a flat-space or quasi-static approximation. We use the curvature-aware Newtonian-gauge EFT equations written by Gambino and Pace [7]. Their metric convention is

$$ds^2 = -(1 + 2\Phi)dt^2 + a^2(1 - 2\Psi)\gamma_{ij}dx^i dx^j,$$

with

$$\delta T^0_0 = -\delta\rho, \quad \delta T^0_i = (\rho + p)D_i v.$$

For the present cubic-Galileon subclass,

$$\alpha_M = \alpha_T = \alpha_H = 0, \quad M^2 = M_{\text{Pl}}^2.$$

The curvature-aware equations use the Gleyzes/Gambino–Pace braiding normalization. To distinguish it from the Bellini–Sawicki braiding function used in the previous sections, we write

$$\boxed{\alpha_B^{\text{GP}} = -\frac{1}{2}\alpha_B^{\text{BS}}}.$$

The kineticity is unchanged,

$$\alpha_K^{\text{GP}} = \alpha_K^{\text{BS}},$$

and therefore

$$\alpha_K + 6(\alpha_B^{\text{GP}})^2 = \alpha_K + \frac{3}{2}(\alpha_B^{\text{BS}})^2 = D_{\text{kin}}.$$

For ideal dust and ideal radiation we set the scalar anisotropic stress to zero, so the traceless spatial equation gives

$$\boxed{\Phi = \Psi}.$$

On the physical $n = 2$ harmonic,

$$D^2 Q = -8KQ, \quad (D^2 + 3K)Q = -5KQ.$$

Writing

$$\kappa = \frac{K}{a^2},$$

the sourced 00 equation becomes

$$6(1 + \alpha_B^{\text{GP}})H\dot{\Phi} + (\alpha_K - 6\alpha_B^{\text{GP}})H^2\dot{\pi} + \left[(6 - \alpha_K + 12\alpha_B^{\text{GP}})H^2 + 10\kappa\right]\Phi$$

$$+6H \left[\frac{\rho+p}{2M_{\text{Pl}}^2} + \dot{H}(1 + \alpha_B^{\text{GP}}) + \kappa \left(1 - \frac{8}{3} \alpha_B^{\text{GP}} \right) \right] \pi = -\frac{\delta\rho}{M_{\text{Pl}}^2}.$$

The $0i$ equation is

$$2\dot{\Phi} - 2H\alpha_B^{\text{GP}}\dot{\pi} + 2(1 + \alpha_B^{\text{GP}})H\Phi + \left(2\dot{H} + \frac{\rho+p}{M_{\text{Pl}}^2} \right) \pi = -\frac{q}{M_{\text{Pl}}^2},$$

where

$$q = \rho_m v_m + \frac{4}{3} \rho_r v_r, \quad \delta\rho = \rho_m \delta_m + \rho_r \delta_r.$$

These equations form a 2×2 local constraint-transfer system for

$$(\dot{\Phi}, \dot{\pi}).$$

Its determinant is

$$\boxed{\det \mathcal{C} = -2H^2 \left[\alpha_K + 6(\alpha_B^{\text{GP}})^2 \right] = \boxed{-2H^2 D_{\text{kin}}}}.$$

Thus the same positive- D_{kin} branch used in the background and Round-3 kinetic audit guarantees that this sourced rate block is nonsingular.

For the velocity convention above, covariant conservation gives

$$\dot{\delta}_m = \frac{k^2}{a^2} v_m + 3\dot{\Phi}, \quad \dot{v}_m = -\Phi,$$

and

$$\dot{\delta}_r = \frac{4}{3} \frac{k^2}{a^2} v_r + 4\dot{\Phi}, \quad \dot{v}_r = -\Phi - \frac{1}{4} \delta_r,$$

with

$$\frac{k^2}{a^2} = 8\kappa.$$

The repository implements these equations as a local linear transfer matrix with the additional topographic response explicitly switched off. This is deliberately not yet promoted to a globally integrated observable solution. The curvature-aware EFT equations and the Round-3 ADM kernel use different perturbation variables and normalizations; the remaining evolution equations must first be matched analytically and then subjected to a Bianchi/constraint-preservation audit. Only after that compatibility gate is passed will we integrate the full matter-era $n = 2$ system and add the independent reduced topographic operator.

8 ADM–EFT Dictionary and Differential-Algebraic Consistency

Rounds 3 and 4 use complementary perturbation languages. Akama and Kobayashi formulate the non-flat Horndeski action in unitary ADM gauge,

$$N = 1 + \delta n, \quad N_i = D_i \chi, \quad g_{ij} = a^2 e^{2\zeta} \gamma_{ij},$$

while Gambino and Pace restore the scalar with

$$t \rightarrow t + \pi$$

and give the curvature-aware field equations in Newtonian gauge [6, 7].

The linear time shift that removes the scalar ADM shift gives

$$\pi = -\chi,$$

$$\Phi = \delta n - \dot{\pi} = \delta n + \dot{\chi},$$

and

$$\Psi = -\zeta + H\pi = -\zeta - H\chi.$$

The inverse dictionary is

$$\chi = -\pi, \quad \delta n = \Phi + \dot{\pi}, \quad \zeta = H\pi - \Psi.$$

8.1 Correct Bianchi audit

The first implementation attempt treated the 00 and 0*i* equations as a complete local evolution system and then demanded that the spatial-trace and scalar equations vanish for an arbitrary pointwise state. That test is too strong and uses the constraint equations in the wrong role.

The correct formulation is differential algebraic. The spatial-trace and scalar equations contain the second derivatives of the physical scalar sector and determine

$$(\ddot{\Phi}, \ddot{\pi}),$$

while the 00 and 0*i* equations define constraints on the initial data. Covariant consistency requires those constraints to be preserved by the evolution.

For

$$\alpha_M = \alpha_T = \alpha_H = 0, \quad \Phi = \Psi,$$

the acceleration matrix is

$$\mathcal{A} = \begin{pmatrix} 2 & -2H\alpha_B \\ 6H\alpha_B & H^2\alpha_K \end{pmatrix},$$

where α_B is in the Gambino–Pace/Gleyzes normalization. Its determinant is

$$\det \mathcal{A} = 2H^2 (\alpha_K + 6\alpha_B^2) = 2H^2 D_{\text{kin}} > 0.$$

Thus the second-order scalar–metric evolution block is nonsingular on the same positive- D_{kin} branch.

The repository now integrates the trace/scalar equations together with ideal dust and radiation conservation, starting from data that satisfy the sourced 00 and 0*i* constraints. The Bianchi audit is then the preservation of those two constraints over the interval

$$2.1 \geq z \geq 0.$$

Both raw and term-normalized constraint residuals are evaluated. The topographic reduced-response operator remains switched off throughout this certification.

9 Selective Harmonic Response

Let

$$L_c(a) = aR_H\ell, \quad L_c^2 p_n = \ell^2 n(n+2),$$

and write

$$K_n = p_n C_n, \quad C_n = c_s^2 - \chi \left[1 - \ell^2 n(n+2) \right].$$

The physical scalar spectrum begins at $n = 2$. Selective softening of $n = 2$ with stiffening of every propagating $n \geq 3$ requires

$$1 - 8\ell^2 > 0, \quad 1 - 15\ell^2 < 0,$$

so

$$\boxed{\frac{1}{15} < \ell^2 < \frac{1}{8}}.$$

Then

$$C_2 < c_s^2, \quad C_n > c_s^2 \quad (n \geq 3).$$

The formal $n = 1$ coefficient also lies on the soft side, but $n = 1$ is not a propagating scalar degree of freedom in the regular one-field closed-FLRW sector.

If $0 < C_2 \ll 1$,

$$T_2 \simeq \frac{S_2}{p_2 C_2}$$

can be enhanced while higher physical harmonics remain regular. For the next physical level,

$$\frac{T_3}{T_2} = \frac{S_3}{S_2} \frac{8C_2}{15C_3},$$

so

$$\boxed{\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}}.$$

For the pure-dipole A_{4i} subspace of the selected $n = 2$ harmonic, the observer-sky pattern is exactly dipolar with geometric radial envelope $\sin 2\psi$.

10 Observable Projection

The observable distance perturbation is not identified directly with the scalar amplitude. At linear order it is a light-cone projection of metric, velocity, and scalar perturbations:

$$\frac{\delta D_L}{\bar{D}_L} = \int d\chi \left[W_\Phi \Phi + W_\Psi \Psi + W_v v_\parallel + \dots \right].$$

If the selected $n = 2$ curvature-scale sector admits a physical gauge-invariant metric response, its exact scalar angular dependence can factor as

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{p}},$$

leading schematically to

$$\frac{\delta D_L}{\bar{D}_L} = \epsilon_D(z) \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}.$$

The existence and normalization of $\epsilon_D(z)$ are therefore bridge outputs to be derived, not consequences of the scalar harmonic geometry alone.

For supernovae,

$$\mu = 5 \log_{10} D_L + \text{const},$$

and therefore a small magnitude dipole A_μ corresponds to

$$\frac{\delta D_L}{D_L} = \frac{\ln 10}{5} A_\mu \cos \theta.$$

For structure growth we define

$$k^2 \Psi = -4\pi G a^2 \mu(k, a) \rho_m \delta_m, \quad \Phi = \gamma(k, a) \Psi,$$

and

$$\Sigma(k, a) = \frac{\mu(k, a)}{2} [1 + \gamma(k, a)].$$

The matter growth factor obeys

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_m(a) \mu(k, a) D = 0,$$

with prime denoting $d/d \ln a$. The observable is

$$f \sigma_8(k, z) = \frac{D'}{D} D \sigma_{8, \text{ini}} = D'(k, z) \sigma_{8, \text{ini}}.$$

The quasi-static limit is used only on scales inside the scalar sound horizon [8]. Horizon-scale $n = 1$ observables require the full constrained long-wavelength response and must not be obtained by extrapolating quasi-static formulas into the first physical curvature-scale mode.

11 Predictions and Falsification Criteria

11.1 Low-redshift calibration

The Pantheon+ data release and cosmological analysis provide the supernova basis for the carried-forward tomographic calibration used here [9, 10]. The earlier scalar-topographic analysis reported

$$A_\mu^* = -0.0412 \pm 0.009$$

in the bin

$$0.01 < z < 0.03,$$

with best-fit direction

$$(\text{RA}, \text{Dec}) = (211.48^\circ, -12.81^\circ).$$

This corresponds to

$$\frac{\delta D_L}{D_L} = (-0.01895 \pm 0.00414) \cos \theta,$$

and at sufficiently low redshift,

$$\frac{\delta H}{H} = (+0.01895 \pm 0.00414) \cos \theta.$$

These numbers are a carried-forward calibration input; the present repository does not independently rerun the Pantheon+ tomographic likelihood. They normalize only the observed $n = 1$ response,

$$A_\mu = \frac{\mathcal{K}_{\text{SN}} S_2}{p_2 C_2},$$

and do not independently determine S_2 , $\mathcal{K}_{\text{SN}}$, or C_2 . Their interpretation as a metric/light-cone response is additionally conditional on resolving the exceptional closed-FLRW $n = 1$ sector.

11.2 Curvature

The pre-existing prospective curvature forecast is retained without retuning:

$$\boxed{\Omega_k = -0.018 \pm 0.004}.$$

At $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$R_H = \frac{c}{H_0 \sqrt{|\Omega_k|}} \simeq 33.15 \text{ Gpc}.$$

The sign prediction is $\Omega_k < 0$ in the usual FLRW curvature convention.

This value is a frozen model forecast, not a current empirical best fit. DESI DR2 BAO results are reported as well described by flat Λ CDM, even while combinations with other probes motivate tests of extensions such as dynamical dark energy [11]. The purpose of retaining -0.018 ± 0.004 is therefore prospective falsification: it must be compared against independent curvature inference without being moved toward the observed result.

11.3 Harmonic hierarchy

The bridge predicts selective $n = 2$ softening with no analogous $n \geq 3$ enhancement. A future source-normalized S^3 $n = 2$ susceptibility comparable to or larger than the $n = 1$ susceptibility would reject the minimal harmonic-filter branch. This statement is made at the compact-space harmonic level; comparison to a measured sky quadrupole requires an explicit observer-shell projection.

11.4 Fixed-axis test

A prospective independent supernova sample should first test the carried-forward axis

$$(211.48^\circ, -12.81^\circ)$$

rather than scanning freely over the sky. A free-axis search may be reported secondarily.

11.5 High- n recovery

At large n ,

$$C_n \simeq \chi \ell^2 n^2,$$

so the extra scalar susceptibility falls as n^{-2} . The model therefore predicts a strong separation between the first nonconstant compact-space level and conventional clustering modes.

11.6 Failure conditions

The minimal bridge is disfavored if one or more of the following are established with adequate precision:

1. independent curvature measurements exclude the frozen negative- Ω_k forecast;
2. the source-normalized S^3 $n = 2$ response is comparable to or exceeds $n = 1$;
3. an independent low-redshift sample excludes the frozen dipole axis while finding a stable incompatible physical axis;
4. a coherent metric/light-cone dipole of comparable amplitude persists into a redshift regime in which the completed model predicts loss of enhanced susceptibility;
5. ordinary RSD modes show a large scale-independent modification incompatible with the high- n suppression;
6. lensing requires a metric response incompatible with the same completed scalar-metric propagator that controls matter clustering;
7. no gauge-invariant physical metric response exists for the exceptional closed-FLRW harmonic corresponding to the proposed $n = 1$ topographic mode.

12 Numerical Implementation and Preregistration Protocol

The numerical pipeline is separated into

$$\boxed{\text{background} \rightarrow \text{physical scalar response} \rightarrow \text{observables} \rightarrow \text{comparison.}}$$

Prediction generation and data comparison are implemented in separate scripts.

12.1 Background system

Let $v = \dot{T}$. Expanding the scalar current equation gives

$$\left(1 + \frac{12Hv}{\Lambda^3}\right) \dot{v} + \frac{6v^2}{\Lambda^3} \dot{H} = \mu^3 - 3H \left(v + \frac{6Hv^2}{\Lambda^3}\right).$$

The Raychaudhuri equation gives

$$\frac{2v^2}{\Lambda^3} \dot{v} - 2M_{\text{Pl}}^2 \dot{H} = \rho_m + \frac{4}{3}\rho_r + v^2 + \frac{6Hv^3}{\Lambda^3} - \frac{2M_{\text{Pl}}^2}{a^2 R_H^2}.$$

At each integration step these equations form a 2×2 linear system for $(\dot{v}, \dot{H})$. The remaining equations are

$$\dot{T} = v, \quad \dot{a} = aH.$$

The Friedmann equation is retained as an independent residual check:

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) - (\rho_m + \rho_r + \rho_T) = 0.$$

12.2 Dimensionless variables

We integrate in $N = \ln a$ with

$$E = \frac{H}{H_0}, \quad \phi = \frac{T}{M_{\text{Pl}}}, \quad \lambda = \frac{\Lambda^3}{M_{\text{Pl}} H_0^2}, \quad q = \frac{\mu^3}{M_{\text{Pl}} H_0^2}.$$

The exploratory reference branch R1 uses

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3.$$

The constant V_0 is determined only by the shooting condition $E(a = 1) = 1$.

12.3 Initial conditions

The numerical integration begins at

$$z_i = 100.$$

The decaying homogeneous current solution is removed using the matter-era particular solution

$$J_i \simeq \frac{2\mu^3}{9H_i}.$$

The initial scalar velocity is the positive small-velocity root of

$$v_i + \frac{6H_i v_i^2}{\Lambda^3} = J_i.$$

12.4 Growth grid

Before prospective comparison, predictions are generated on the fixed grid

$$k = \{0.02, 0.05, 0.10, 0.15, 0.20\} \text{ Mpc}^{-1}$$

and

$$z = \{0.5, 0.8, 1.0, 1.5, 2.1\}.$$

A deterministic JSON manifest containing the action version, parameters, numerical tolerances, calibration values, and predicted observables is hashed with SHA-256. Any later scientific or numerical revision creates a new manifest version rather than overwriting the previous freeze.

13 Results: Reference Branch R1

R1 is an exploratory reference branch rather than a uniquely derived BHSM cosmology. Its purpose is to demonstrate that the bridge can simultaneously realize late-time scalar activation, a selectively enhanced $n = 1$ response, and near-standard high- n structure growth.

For

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3,$$

the shooting condition $E(1) = 1$ gives

$$\frac{V_0}{M_{\text{Pl}}^2 H_0^2} = 2.39861.$$

z	α_B	D_{kin}	Ω_T
10	2.07×10^{-7}	1.64×10^{-6}	0.00194
3	8.60×10^{-5}	6.80×10^{-4}	0.03915
2.1	3.72×10^{-4}	0.00294	0.08060
1.5	0.00122	0.00959	0.14321
1.0	0.00380	0.02988	0.24551
0.8	0.00622	0.04889	0.30787
0.5	0.01346	0.10536	0.43200
0.3	0.02271	0.17740	0.53538
0	0.04940	0.38485	0.70791

Table 1: Reference-branch background evolution.

The scalar braiding and no-ghost combination evolve as shown in Table 1.

The scalar equation of state evolves from nearly vacuum-like behavior at high redshift to

$$w_T(z=0) \simeq -0.906,$$

and the background acceleration transition occurs at

$$z_{\text{acc}} \simeq 0.683.$$

Relative to a curved Λ CDM model with identical $H_0, \Omega_{m0}, \Omega_{r0}, \Omega_{k0}$, the R1 expansion rate differs by only order one percent over the late-time interval. The maximum difference in the tabulated range is about 1.7% near $z \sim 0.5$.

13.1 Prospective growth target

In the minimal high- n realization, the direct topographic fifth-force term is negligible on ordinary RSD modes and the dominant growth shift is produced by the altered background history. With identical matter-era normalization,

$$R_{f\sigma_8}(z) = \frac{f\sigma_{8,\text{R1}}}{f\sigma_{8,\Lambda\text{CDM}}}$$

is

The central phenomenological result is therefore not a large generic modification of gravity. It is the combination

enhanced horizon-scale $n = 1$ susceptibility
+ rapid high- n recovery
+ few-percent late-time growth shift

14 Reference-Branch Figures

15 Discussion

The closed-FLRW gauge audit produces a concrete correction. The first nonconstant scalar harmonic in our original notation, $n = 1$, maps to the exceptional $N = 2$ sector of standard closed-universe perturbation theory. Its scalar-derived trace-free tensor harmonic vanishes, and the regular

z	$R_{f\sigma_8}$	suppression
2.1	0.99346	0.65%
1.5	0.98869	1.13%
1.0	0.98164	1.84%
0.8	0.97795	2.20%
0.5	0.97271	2.73%
0.3	0.97168	2.83%
0.0	0.98482	1.52%

Table 2: R1 growth forecast relative to matched curved Λ CDM. These values are reference-branch predictions and become formally preregistered only when reproduced by the canonical code and written to the hashed manifest.

generally covariant one-field scalar sector contains no propagating scalar degree of freedom there. Modern closed-inflation calculations likewise take physical scalar modes to begin at $N \geq 3$ [2, 3].

The natural response is to move the topographic carrier to the first physical scalar level: our $n = 2$, equivalent to $N = 3$. A degree-two harmonic on S^3 is a traceless quadratic form in the $\mathbb{R}^4$ embedding coordinates. Its cross components A_{4i} generate an exact observer-sky dipole

$$T_{2,\text{dip}} \propto \sin 2\psi (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})$$

without requiring a sky quadrupole.

The reduced-response filter also adapts naturally. The corrected window

$$\frac{1}{15} < \ell^2 < \frac{1}{8}$$

softens $n = 2$ while stiffening every higher propagating scalar harmonic $n \geq 3$. The formal $n = 1$ coefficient is irrelevant to the regular physical scalar spectrum. The enhanced carrier is therefore the first physical compact-space scalar mode.

The role of BHSM remains structural rather than a completed microscopic derivation. The imported element is the constraint-reduced response logic. The specific mixed block and the source law S_2 remain exploratory until derived from the parent boundary action.

The complete $n = 2$ scalar-metric transfer kernel must still be derived in the closed cubic-Galileon background. $D_{\text{kin}} > 0$ remains an inherited kinetic diagnostic rather than a complete closed-background stability proof. R1 values $q = 3$ and $\lambda = 1$ remain reference choices, and the frozen curvature forecast must not be retuned to observational results.

The R1 numerical background and high- n growth forecast do not depend on this harmonic correction. Their frozen manifest is therefore preserved unchanged, while the theoretical amendment is recorded separately before comparison with the targeted independent datasets.

16 Conclusions

The closed-FLRW gauge analysis resolves the principal ambiguity in the original bridge construction. Our $n = 1$ scalar harmonic maps to the exceptional $N = 2$ sector and is not retained as the ordinary propagating scalar carrier. The first physical scalar level is

$$\boxed{n_\star = 2} \quad (\boxed{N_\star = 3}).$$

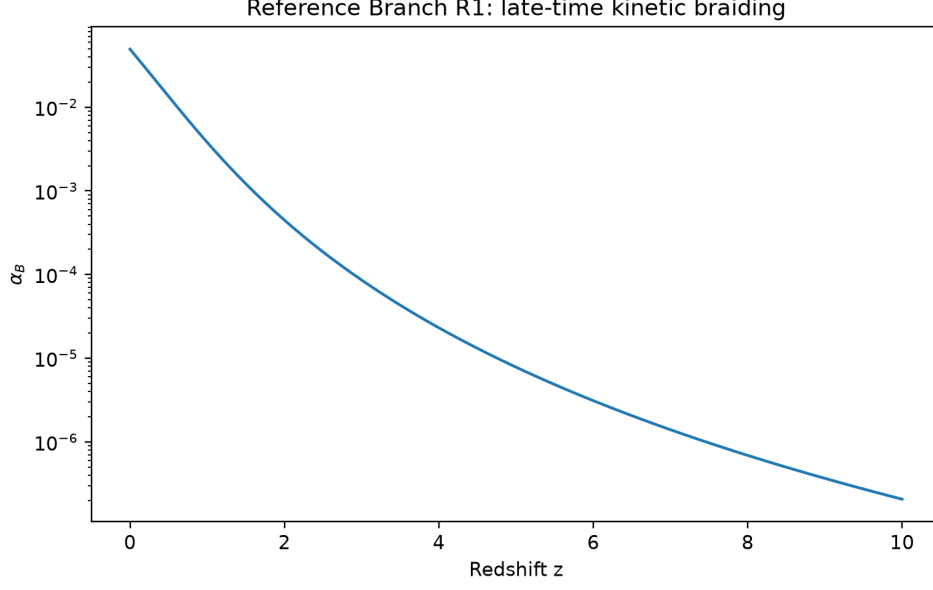


Figure 1: Kinetic-braiding function $\alpha_B(z)$ for Reference Branch R1. The logarithmic vertical scale emphasizes the suppression of the braiding sector toward high redshift.

This first physical harmonic contains an exact sky-dipole subspace,

$$T_{2,\text{dip}} = A \sin[2\psi(z)](\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}),$$

when only the cross components A_{4i} are populated.

The corrected selective-response window is

$$\boxed{\frac{1}{15} < \ell^2 < \frac{1}{8}},$$

within which $n = 2$ is softened and every higher propagating mode $n \geq 3$ is stiffened. The next-level source-normalized response obeys

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}.$$

The carried-forward supernova amplitude now calibrates the physical $n = 2$ dipole response. The R1 background and high- n growth predictions remain unchanged.

The revised signature is therefore

$$\boxed{\begin{aligned} &\text{first physical } S^3 \text{ scalar mode } (n = 2) \text{ enhanced} \\ &\oplus \text{ fixed-axis dipole subspace with } \sin 2\psi \text{ envelope} \\ &\quad \oplus \text{ suppressed } n \geq 3 \text{ susceptibility} \\ &\quad \oplus \text{ near-GR high-}k \text{ growth.} \end{aligned}}$$

The next calculation is the full gauge-invariant $n = 2$ scalar-metric transfer matrix in the closed cubic-Galileon background.

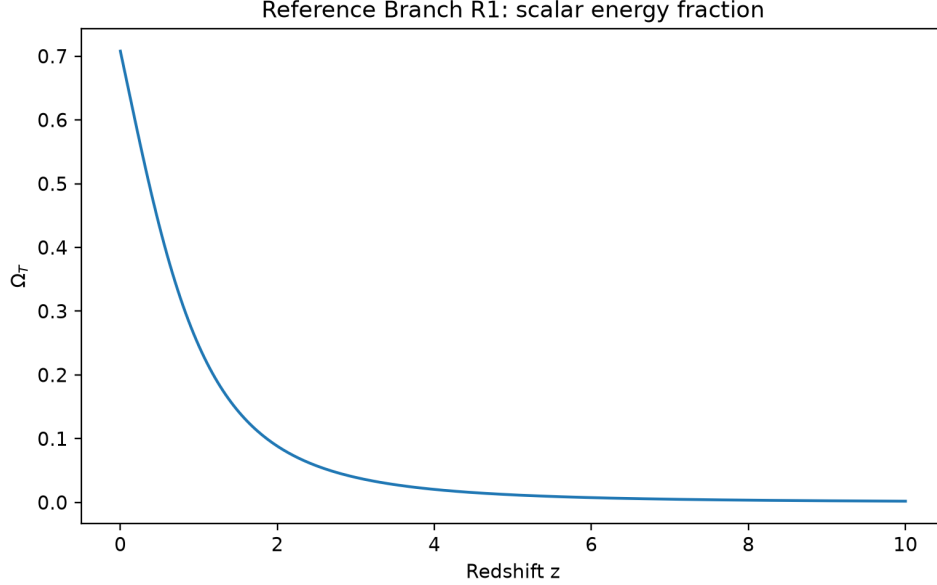


Figure 2: Scalar energy fraction $\Omega_T(z)$ for R1. The scalar component is subdominant at high redshift and becomes dynamically important only at late times.

References

- [1] Norman P. Carberry. A hyperspherical scalar–topographic framework for late-time cosmological anomalies. Preprints.org, 2026. Posted 20 January 2026, doi:10.20944/preprints202601.1427.v1.
- [2] Claus Kiefer and Tatevik Vardanyan. Power spectrum for perturbations in an inflationary model for a closed universe. *General Relativity and Gravitation*, 54:30, 2022. doi: 10.1007/s10714-022-02918-3.
- [3] Enrico Specogna, Tatevik Vardanyan, William Giarè, and Eleonora Di Valentino. Slow rolling down the curvature: A reassessment of the planck constraints on ϕ^2 inflation in a closed universe. *Physical Review D*, 113:123526, 2026. doi: 10.1103/dy2d-z3cc.
- [4] Cedric Deffayet, Oriol Pujolas, Ignacy Sawicki, and Alexander Vikman. Imperfect dark energy from kinetic gravity braiding. *JCAP*, 10:026, 2010. doi: 10.1088/1475-7516/2010/10/026.
- [5] Emilio Bellini and Ignacy Sawicki. Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity. *JCAP*, 07:050, 2014.
- [6] Shingo Akama and Tsutomu Kobayashi. General theory of cosmological perturbations in open and closed universes from the horndeski action. *Physical Review D*, 99:043522, 2019. doi: 10.1103/PhysRevD.99.043522.
- [7] Serena Gambino and Francesco Pace. Linear perturbations in horndeski theories with spatial curvature, 2024. Proceedings contribution, MG17.
- [8] Ignacy Sawicki and Emilio Bellini. Limits of quasistatic approximation in modified-gravity cosmologies. *Phys. Rev. D*, 92:084061, 2015.

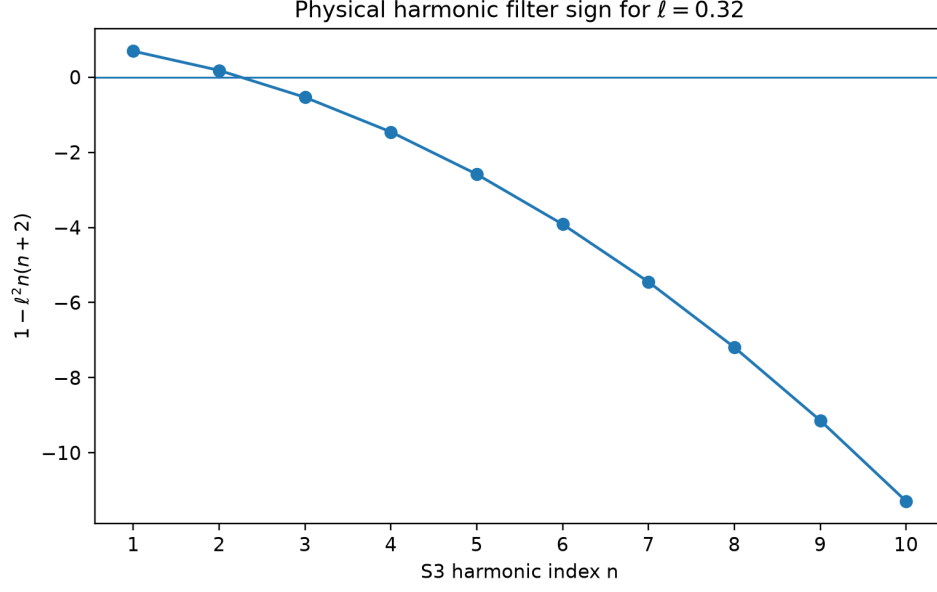


Figure 3: Sign of the Schur-response factor $1 - \ell^2 n(n+2)$ for the illustrative filtering value $\ell = 0.40$. The $n = 1$ branch lies on the softening side while every displayed $n \geq 2$ mode lies on the stiffening side.

- [9] D. Scolnic et al. The pantheon+ analysis: The full dataset and light-curve release. *Astrophys. J.*, 938:113, 2022.
- [10] D. Brout et al. The pantheon+ analysis: Cosmological constraints. *Astrophys. J.*, 938:110, 2022.
- [11] DESI Collaboration. Desi dr2 results ii: Measurements of baryon acoustic oscillations and cosmological constraints. *Physical Review D*, 112:083515, 2025.

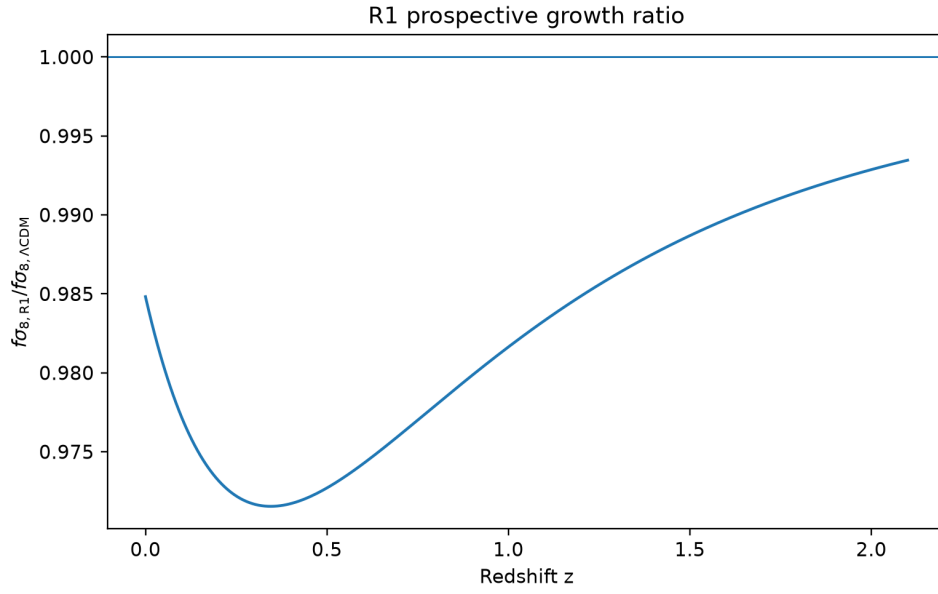


Figure 4: Prospective R1 growth ratio relative to a matched curved- Λ CDM background with the same early normalization.